SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

(Deemed to be University)

Department of Mathematics and Computer Science

SYLLABUS: USTA-201: Probability Theory and Distributions

Course Code	USTA-201	Credits: 4 (3-1-0)	Total Periods: 56
Prerequisites	Introductory Statistics, Basic calculus		

Course Objectives

The main objective of this course is to provide students with the foundations of probabilistic and statistical analysis. The topic covered will include probability theory, random variable and limit theory. The course also focuses on the understanding of discrete and continuous distributions with major emphasis on Bernoulli, Binomial, Poisson, Geometric distributions, Uniform, Cauchy, Normal, Exponential, Gamma distributions.

Course Outcomes

At the end of the course, students will be able to

1. Explain the classical, empirical and axiomatic approaches to probability, and compute probabilities using conditional probability, independence and Bayes' theorem.
2. Define discrete and continuous random variables and determine their probability distributions and distribution functions.
3. Compute expectation, variance, moment generating functions and characteristic functions of random variables.
4. Derive the properties of the Bernoulli, binomial, Poisson and geometric distributions and apply them to real-world problems.
5. Derive the properties of the uniform, Cauchy, normal, exponential and gamma distributions and apply them to real-world problems.
6. Apply Chebyshev's inequality, the laws of large numbers and the central limit theorem to approximate probabilities and justify statistical methods.

Syllabus

Unit	Topic	Contents	Periods
1	**Probability**	Random phenomena – Basic concepts of Probability – Classical Probability – Empirical Probability – Axiomatic approach to Probability – conditional probability – marginal probability – Independence of events – Bayes theorem.	10
2	**Random Variables**	Events and random variables – Discrete and continuous random variables – Probability distribution – Distribution function – Mathematical expectation – variance – Moment generating and characteristic functions.	12
3	**Discrete Distributions**	Bernoulli, Binomial, Poisson, Geometric distributions, Properties and applications.	12
4	**Continuous Distributions**	Uniform, Cauchy, Normal, Exponential, Gamma distributions – Properties and applications.	12
5	**Limit theorems**	Types of convergence, Chebyshev's inequality, Weak and strong laws of large numbers – De Moivre's – Laplace central Limit Theorem.	10

Unit	Topic	Contents	Periods
		Total	**56**

Key Text(s) / Readings

- S.C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand & Sons, 2014.
- J. E. Freund, Mathematical Statistics, Prentice Hall of India., 2013, eighth edition.
- Ross S, A First Course in Probability, Pearson Education., 2013, ninth edition. (Chap. 1-5,7,8)
- J.L. Devore, Probability and Statistics for Engineering and Sciences, Brookes / Cole Publishing Co., 2017, ninth edition.